PAARTH KAPUR

Linkedin :www.linkedin.com/in/paarth-kapur- 7b352a201
+91 9167849865
paarthkapur@gmail.com

Work Experience

Business Analyst *Sep 2024 - Current*
Tata Play Pvt Ltd (A.R. Chawra Associates – 3P Payroll) - Mumbai

- Lead execution of the profitability model, managing SQL scripts, PostgreSQL databases, and Spoon (Pentaho) workflows to drive business insights.
- Design and deliver business-critical reports, including a daily subscriber burn rate tracker, enhancing data visibility and decision-making.
- Identify and resolve revenue leakages, ensuring data accuracy and optimizing business processes for efficiency.
- Automated monthly reporting, reducing time and effort by **75%**, and implemented high-volume data entry automation using Python & Selenium, cutting processing time by **83%**.
- Utilized Python for large-scale data handling and developed automation solutions for repetitive workflows.
- Applied NLP techniques to classify customer reviews, enhancing sentiment analysis and actionable insights; developed models to segment reviews by sentiment and issue type.
  Tools & Technologies: Power BI, Power Automate, SQL, PostgreSQL, Excel, Python, Selenium, Spoon (Pentaho)

Data Analyst *Jan 2024 - Jun 2024*
Colgate Global Business Solutions - Mumbai

- Key member of the APAC Scrum Team, leveraging **Snowflake SQL** to optimize processes across Material Master, Sales, Stock Management, and Forecasting.
- Ensured accuracy of supply chain reports by validating KPIs, reconciling cross-system data, investigating discrepancies, and managing monthly refresh cycles for reliable decision-making.
- Designed and delivered a **Snowflake-powered Stock Availability Report** integrating supply–demand logic, safety stock thresholds, lead times, and country-specific calculations, enabling informed regional decisions.
- Built a **low-code Outsystems prototype** connecting SAP to an external webpage, enhancing operational data accessibility.
- Provided technical support through test script preparation, test data management, and automation issue resolution.
  Tools: Snowflake (SQL), DOMO, Sigma, Excel (Advanced), Airflow, DBT, SAP, Outsystems, GitHub

Programmer Analyst *Jun 2023 - Jul 2023*
Mobicule Technologies Netrascan - Mumbai

- Developed a **Python + Flask application** using CNN models to detect and mask sensitive information in images, adaptable to varied orientations.
- Executed the **full DevOps lifecycle** from planning to deployment, ensuring a streamlined and efficient workflow.
- Built **Python scripts for data pipelining, cleaning, and automation**, improving processing efficiency and significantly reducing manual effort.
  Technologies: Python, Flask, Deep Learning (CNN), Microsoft Excel, Data Analysis

Technical Intern – ArcGIS & Machine Learning | *Sep 2022 - Apr 2023*
K. J. Somaiya College of Engineering - Mumbai

- Processed **Landsat 8 OLI/TIRS C2 L1 TIF files** and applied **NDVI** to analyze vegetation density and health.
- Implemented **remote sensing classification techniques** to differentiate wetland and dryland ecosystems using ArcGIS.
- Performed **accuracy assessment with stratified random sampling**, achieving a classification accuracy of **89.42%**.
- Demonstrated the potential of **geospatial analysis** for urban planning, land cover mapping, habitat classification, and spectral correlation.

Technical Intern – Deep Learning *Jul 2022 - Apr 2023*
K. J. Somaiya College of Engineering - Mumbai

- Processed thermal images using **color transformations, cropping, and thresholding** to enhance data quality for analysis.
- Applied **image segmentation** with HSV thresholds, masking, and bounding boxes to extract regions of interest.
- Built and trained a **CNN model** with convolutional, pooling, and dense layers for thermal image classification.
- Leveraged **ResNet50** as a feature extractor with Region Proposal Network (RPN) and ROI pooling for precise region detection and analysis.
- Evaluated model predictions using **thresholding, IoU calculations, and segmentation metrics**, improving detection accuracy.

Technical Intern – ArcGIS Data Exploration *Jan 2023 - Feb 2023*
K. J. Somaiya College of Engineering - Mumbai

- Processed and standardized **~950 GPX files into SHP format** using QGIS, consolidating shapefiles across two districts for improved data usability.
- Refined **field boundary accuracy** via Google Earth Pro and performed **KML-to-SHP conversions** for geospatial consistency.
- Analyzed **Landsat 8 satellite imagery** to compute NDVI, generating actionable insights on vegetation density and supporting precise land cover classification.

Technical Intern – Data Science *Jul 2021 - Nov 2021*
K. J. Somaiya College of Engineering - Mumbai

- Created **clear and effective visualizations** using Python libraries to communicate insights from complex datasets.
- Applied **data sorting and preprocessing techniques** for efficient analysis and organization.
- Worked with **Neo4j graph database** to store and query website connectivity data.
- Built **interactive visualizations using Bokeh** to enable dynamic connectivity graph analysis.

Education

Bachelor of Technology in Information Technology : *Honors in Artificial Intelligence* *May 2024*
K.J. Somaiya College of Engineering

- **CGPA**: 9.64/10

Publications

- *Transforming College Counseling: An In-House Solution for Student Mental Wellness*. **IEEE**, Mar 2023
  Designed a mobile and web application with an intuitive interface to promote student mental wellbeing.
- *Leveraging Swin Transformer for Robust Object Detection in Autonomous Vehicles*. **IEEE**, Jul 2025
  Demonstrated superior accuracy and robustness of Swin Transformer over CNN-based detectors under adverse driving conditions, while maintaining real-time efficiency.
- *Lung Cancer Detection Using Medical Image Processing*. **Springer**, Oct 2025
  Developed a hybrid ResNet50–DenseNet201 model for lung cancer classification using CT scans, benchmarked against EfficientNet (94.29% accuracy), enhancing early detection capabilities.
- *Towards Explainable Neural Networks: Investigating Image Pixel Contributions with LRP and Explainable AI*. **Springer**, Oct 2025
  Implemented a custom LRP approach with VGG16 to improve CNN interpretability via pixel-level visualizations, integrating real-time video preprocessing for greater transparency in Explainable AI systems.

Certificates

- IBM Data Analytics with Excel and R Specialization | IBM
- NoSQL, Big Data, and Spark Specialization | IBM
- Swift 5 iOS Application Developer Specialization | LearnQuest
- Harnessing the Power of Data with Power BI | Microsoft
- Preparing Data for Analysis with Microsoft Excel | Microsoft
- Graph Analytics for Big Data | UC San Diego
- Firm Level Economics: Consumer and Producer Behavior | University of Illinois Urbana-Champaign
- Big Data Modeling and Management Systems | UC San Diego
- ArcGIS Training | IEEE GRSS

Volunteering Experience

- **Utkarsha Mandal | Volunteer Trainer | Apr 2024 – Dec 2024**
  Conducted training sessions on Microsoft Excel (formulas, data organization, visualization) and Python basics, improving students' analytical and coding skills. Contributed to youth empowerment through technology-driven learning initiatives.
- **Rotary Club of Bombay Seacoast | Volunteer (Animal Shelter) | Jun 2024 – Sep 2024**
  Applied data analysis to monitor adoption rates, maintain medical care records, and assess resource needs for rescued animals. Streamlined operations, enhancing efficiency and overall animal welfare.
- **NGO Cell Mumbai | Volunteer (Healthcare Initiatives) | May 2024 – Aug 2024**
  Assisted in organizing healthcare camps (e.g., eye check-ups), coordinating patients, and managing event logistics. Supported efficient delivery of medical services to underserved communities.
- **Vile Parle Cultural Center | Volunteer (Event Management) | Dec 2018 – Dec 2024**
  Coordinated logistics and communication for large-scale annual cultural programs. Facilitated smooth event execution through teamwork and stakeholder collaboration.
- **Nurturing Lives | Social Media Volunteer | Apr 2021**
  Managed social media campaigns for *Project Ekasaathe*, creating content to raise awareness about transgender community issues. Boosted engagement and outreach for advocacy initiatives.

Skills

- **Programming & Databases:** SQL, Python, R, NoSQL, relational databases
- **Data Visualization & Analysis:** Power BI, Tableau, Excel (advanced), data reporting & dashboards
- **Business & Automation Tools:** Business Intelligence, automation workflows, low-code/no-code technologies
- **Artificial Intelligence & Analytics:** Machine learning, predictive analytics, statistical modeling
- **Productivity & Reporting:** Report generation, process optimization, data-driven decision-making

Academic Projects

- **Colour Quantization in Images using K-Means | Sep 2022**
  Implemented K-Means clustering to extract predominant color palettes from images and generate quantized versions with reduced colors. Applied Wu's algorithm for deterministic initialization and nearest-neighbor interpolation for efficient downsampling.
- **Customer Segmentation Dashboard | Sep 2023**
  Applied K-Means clustering for customer segmentation using machine learning. Performed data preprocessing, model training, and segment profiling. Developed insights to support personalized marketing strategies based on customer characteristics.

Positions & Responsibilities

- **Head of Marketing | Emfinty | Aug 2022 – Aug 2023**
  Led the marketing team, providing strategic direction and overseeing campaigns aligned with organizational objectives. Drove lead generation and successful deals through effective marketing strategies and team coordination.
- **Event Team Member | Emfinty | Aug 2021 – Aug 2022**
  Planned and organized club events, managing logistics including venue, equipment, and vendor coordination. Created event schedules and reports, ensuring smooth execution through effective team communication and collaboration.